

6809 Forth — Open Items Checklist (Part 3: ANS Test Suite Era)

This is **part 3**, picking up where part 2 ends. Part 2's last entry is the

`ECHOEMIT` fix and the `BASECODE` / `BASEDICT` shift to `$DE5E` / `$D673` .

Parts 1 and 2 are kept unchanged as project history. Where something below overturns an entry in those files, it says so under "Earlier entries superseded".

Sources for this part: the current `forth6809.asm` in the Project (the baseline that assembles and passes all 17 ANS test sections), the ANS Test Suite `README.md` and `bug_fixes.md` , and the session notes kept while fixing the defects listed here. Status marks:

- `[x]` resolved and confirmed by a real test run (MAME and/or minicom).
- `[ ]` open.
- *Not re-assembled here* means the figure comes from reading the source, not from a fresh `lwasm` listing. No assembler is available in the environment this file was written in.

Changes since part 2 — serial I/O and dictionary

- ☒ **`IRQH` rewritten for symmetry between its RX and TX halves, with a single exit point, and receiver errors are now counted.** Framing, overrun and parity errors increment `FECOUNT` , `OVRCOUNT` and `PECOUNT` (page-zero cells), cleared by `INITSERIAL` at cold and warm start. The polling `KEY` counts the same three errors too (only the interrupt path checked them before). A new `POLLREADYCNT` counts polling-mode `KEY` calls that found a character already waiting, as a backlog signal distinct from a true overrun. `BASECODE` and `BASEDICT` shifted down `$30` (48 bytes) to make room.
- ☒ **Polling-mode software flow control added: `PUTXON` / `PUTXOFF` .** `ACCEPT` sends XON at the start of each line and XOFF the moment a line is complete (CR seen), before interpretation or compilation begins, because a polling build has no ring buffer to absorb a host that keeps sending. The bytes go straight to the ACIA without

touching the data stack. `XONCH = $11` , `XOFFCH = $13` . See the open item under "Carried forward" for what is still not implemented.

- ☒ **Dictionary headers added for words that already had code or were new:** `:NONAME` (header added, `BASECODE` shifted), `>NUMBER` (code existed and worked but had no header and was unreachable by name; found by an audit), and `[COMPILE]` (new word `XCOMPILE` , needed for Annex F F.6.2.2530). `[COMPILE]` always compiles a direct call regardless of the `IMMEDIATE` flag; unlike `POSTPONE` it is wrong for default-compilation words such as `CREATE / DOES>` words, which is why `[COMPILE]` is obsolescent. `BASECODE` shifted up `$0C` and `$0E` for the `>NUMBER` and `[COMPILE]` headers respectively, following the 5-bytes-per-entry overhead formula (the earlier 4-byte estimate had caused a 3-byte overflow).

- ☒ **Current memory-map anchors (read from the source's `EQU` lines).**

`INOUT $C000 - $C0FF (ACIA at INOUT+8 )`, `USROMSTRT $C100` ,
`BASEDICT $D543` , `BASECODE $DD54` , `INITCODE $FFA4` ,
`VECTORS $FFF0` . **RAM:** `RSTACK` occupied `$BD00 - $BFFF` , `DSTACK`
top `$BCFF` (1024 bytes) and `CODETOP $B900` , `APPCODE $7000` ,
`APPDICT $2000` , `APPVARS $021B` ending at `APPVARSEND =`
`APPDICT-1 ( $1FFF )`, `TIBBUF $0106` (80 bytes), serial buffer
block from `$0176 / $0180` . *Not re-assembled here.* The `ClaudeForth`
document's Memory Map section still shows earlier values (see
"Documentation to refresh").

Changes since part 2 — defects found and fixed by testing

Each was found by the unit tests, the ANS Annex F tests, or a real run.

`bug_fixes.md` in the ANS Test Suite directory records the last five.

- ☒ **EXIT inside a CASE / OF clause hung in MAME.** The compile-time scan for enclosing structures steps in fixed 4-byte strides, but `CASEW` left a lone 2-byte `TAGCASE` cell, so the scan ran past `CSP` forever. `CASEW` now leaves a 2-cell frame (filler + `TAGCASE`), and `ENDCASE` 's `ECDONE` path pops the paired filler.
- ☒ **+LOOP had three separate bugs in `DOPLUSTEST` .** (1) An extra test exited one pass early whenever the new index landed exactly on the limit with a step other than +1; removed. (2) `LEAVE` -triggered exits skipped popping the step value, leaving a stray stack item;

the pop now happens unconditionally first. (3) The sign-flip crossing test is wrong in general because a signed 16-bit difference changes sign at two points on the 65536-ring; replaced by an unsigned-wraparound test (step>0: crossed iff new<old unsigned; step<0: crossed iff new>old). Verified by simulation and by Annex F section 12.

- ☒ **RECURSE inside :NONAME resolved to the wrong word.** It found "the word being compiled" by walking `LATEST`, but `:NONAME` never updates `LATEST`. Added `CURXT` (an alias of the `NEWHDR` cell, since page zero is full) which `:` and `:NONAME` both set to their own execution token; `RECURSE` reads it directly.
- ☒ **UDIV32 mishandled a 17th remainder bit.** `UM/MOD` with an unsigned divisor near `$FFFF` (Annex F `MAX-UINT MAX-UINT UM* MAX-UINT UM/MOD`) needs a transient 17th bit that the final `ROL DIVREM` produced in carry but never tested. Fixed with a `BCS` test after that rotate. Only `UM/MOD` can reach it; the signed-divisor words cannot. Regression test `TSTUMSM2` added to `unit_tests.asm`.
- ☒ **WORD, PARSE and PARSE-NAME treated >IN beyond the end of input as a huge remaining length** (unsigned wrap), parsing stale memory. Annex F's `>IN` tests set it past the end deliberately. All three now check for borrow and treat overshoot as an exhausted parse area.
- ☒ **2@ / 2! / 2CONSTANT cell order.** The standard puts x2 at the address and x1 at address+cell. Annex F section 18 failed until `DFETCH`, `DSTORE` and `TWOCONSTANT` were made to match. This overturns the earlier part 2 entry that "fixed" `2@` the other way.
- ☒ **CATCH / THROW did not preserve the input source.** A throw out of a nested `EVALUATE` lost the rest of the calling line and left stray data-stack cells (four after section 11). The `CATCH` frame now also saves `SRCADDR`, `SRCLen`, `SRCID` and `>IN`, and `THROW` restores them.
- ☒ **EVALUATE kept the outer source in global EVSAVE* cells, which nesting clobbered.** It now saves and restores them on the return stack (the `EVSAVE*` cells are unused).
- ☒ **ABORT did not throw**, so `CATCH` could not intercept it. It is now `-1 THROW (ABORTW)`; the top level resets silently on `-1`. `COLD` ends with `BRA ABORT` to keep its fall-through to the reset

entry.

- ☒ **UNLOOP and EXIT loop-frame handling.** A `DO` frame on the return stack is 6 bytes (index, limit, `LEAVE` flag); `EXITUNLOOP` discarded 8 per enclosing loop and `UNLOOP` did nothing. `UNLOOP` now drops the frame, and `EXIT` no longer discards frames (the standard: use `UNLOOP` first). Found by Annex F test `GD6` in section 12.

Changes since part 2 — test infrastructure

- ☒ **All 17 ANS Annex F test sections (`ans_tests/08` to `26`) pass with clean data stacks** on MAME (Python runner) and via minicom pasting, after the fixes above. Section files are independent (`MARKER Mnn`), adapted for 16-bit cells where Annex F assumes 32, and corrected where the published test text is wrong.
- ☒ **TEST-BEGIN / TEST-END** (in `00_test_prelude.fs`) give every test file its own report: "TEST SUMMARY: N run, M failed" plus a "STACK IMBALANCE" check. `TEST-REPORT` was removed from `ttester.fs`. `ans_test_runner.py` now reads those lines instead of sending its own reporting commands. `apply_test_markers.py` adds both calls to new section files.
- ☒ **Assembly unit tests (`unit_tests.asm`) re-verified; all pass.** Four tests had outdated expectations after the fixes above and were corrected (the asm was right in each case):
 - `TST2FETCHSTORE` (now expects x2 at the lower address, x1 at `addr+cell`), `TSTRECUR` (sets and restores `CURXT` instead of faking a header via `LATEST`), `TSTUNLOOP` (pushes a fake 3-cell `DO` frame on S and checks S rises by 6 and U is untouched; the old version treated `UNLOOP` as a bare RTS and crashed the CtrlFlow run) and `TSTEXIT` (calls `UNLOOP` before `EXIT` ; the compiled body is now `DO I 3 = IF UNLOOP EXIT THEN I + LOOP`).
- ☒ **Documentation:** ANS Test Suite `README.md` (how to run, edit and read the tests, differences from Annex F, known limitations) and `bug_fixes.md` (history of the defects above).

Earlier entries superseded

- Part 2, " `UNLOOP` is now a true no-op": no longer true. `UNLOOP` drops the loop frame and `EXIT` does not (see above). The `ClaudeForth` glossary

entries for `UNLOOP` and `EXIT` need to follow.

- Part 2, " `2@` read order fixed to low-then-high": reversed again (see above).
- Part 2, "Software (XON/XOFF) handshaking is still not implemented": partly superseded (see below).
- The top-of-file comment in `forth6809.asm` still says "220 entries", still says XON/XOFF "remains not implemented", and still describes a 19-byte `BASECODE` / `BASEDICT` overlap. All three are stale. The source has 227 `H_` header labels; the overlap was closed by later `BASECODE` / `BASEDICT` moves. Refresh that comment block (see below).

Documentation to refresh (not started — awaiting go-ahead)

- ☐ `ClaudeForth.docx` / `ClaudeForth preview.pdf` : Memory Map section (new `BASEDICT` / `BASECODE` / `INITCODE` values and any changed RAM anchors above), the Assembler Source appendix (updated `forth6809.asm`), and the `UNLOOP` / `EXIT` / `2@` / `2!` / `ABORT` glossary entries. Real byte totals and the unused-ROM figure need a fresh `lwasm` listing.
- ☐ `forth6809.asm` top-of-file comment block (items listed above).

Carried forward from part 2 (still open)

- ☐ **Software flow control is only half in place.** Polling builds now transmit XON/XOFF around each line. Nothing recognizes an incoming `$11` / `$13` from the host, and interrupt-driven builds still rely on hardware RTS/CTS only.
- ☐ **No hard boundary checks** as `DPHERE` / `CODEHERE` / `VARHERE` grow; `UNUSED` and `VUNUSED` only report.
- ☐ **CATCH -wrapped QUIT / INTERPRET rollback is scoped to one input line**, not back to the opening `:` of a multi-line definition.
- ☐ **J does not generalize** past one nesting level; `LEAVE` is only correct when textually inside its loop.
- ☐ **Search-Order word set not implemented** (documented, Section 4.5).
- ☐ **Duplicated logic:** `NUMBERQ` 's inline 32-bit negation versus `MNEG32` ; the self-referential PFA fix repeated in `CONSTANT` , `DEFER` , `2CONSTANT` , `MARKER` . Part 2 deferred both until automated testing existed. That condition is now met (17 ANS sections plus the

unit tests), so refactoring them is possible and can be checked.

- ☐ **Design questions:** range-checking in `ALLOT / VALLOT / PICK / ROLL / BASE ;` the `TIB / SOURCE` distinction; `SWIH` 's placeholder `-99 ; REPLACES / SUBSTITUTE` single-slot only.
- ☐ **DUMP leading-line formatting change** (part 2) was left unchecked there; confirm or close.

New open items

- ☐ **Double-Number word set is partly implemented and has no Annex F tests.** Present: `D+ D- DNEGATE DABS DMAX DMIN M+ S>D D>S D= D< DU< D. D.R 2ROT 2VARIABLE 2CONSTANT` and `123.` literals. Appear to be missing: `D0= D0< D2* D2/ M*/ 2LITERAL 2VALUE`. A double-number test file would be the next ANS section, after those words exist.
- ☐ **Facility word set:** only `KEY?` exists; no tests.
- ☐ **ENVIRONMENT? :** `X:deferred` answers false (not in `ENVTABLE`) although `DEFER / IS / ACTION-OF` exist. The test accepts either answer; answering true would be more accurate.
- ☐ **ABORT is silent** (including `-1 THROW` at the top level), and `ABORT"` message output at the top level is not covered by any test. Decide whether either should print.
- ☐ **Re-assemble and regenerate the listing.** Confirm the current `BASEDICT / BASECODE / INITCODE` boundaries and spare bytes (the source comment records a 9-byte margin as of the `[COMPILE]` header, never re-verified), then update the docx Memory Map from that listing rather than from comments.
- ☐ **Annex F test files that are physical-line sensitive** (sections 19 and 24) must not be reflowed; any tooling that rewraps test files has to respect that (documented in the README).
- ☐ **Page-zero budget is still 256/256.** Any new global (for example, for Double-Number work or more error counting) must reuse a cell or move one out of page zero.